EAST WEST UNIVERSITY

Project Report

Title: **Solar Man**: A Web-Based Solar Plant Installation and Management System

Course title: - Software Engineering

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1: Project Introduction

1.1 Introduction and System Overview

Solar Man is a proposed web-based platform that helps homeowners explore solar products, estimate household solar requirements, request site assessments, receive quotations, purchase equipment, track installation and request maintenance. It connects homeowners, technicians and administrators through a single workflow. Calculator results are preliminary; final design and safety decisions require qualified technician validation.

1.2 Scope and Objectives

In scope: account registration/login and role-based access; product catalog and inventory; solar capacity calculator; direct orders and checkout; site assessment and quotations; technician assignment and installation tracking; maintenance and service history; notifications; AI informational assistant; and admin reports.

Out of scope for the first release: live hardware telemetry/control, safety certification by the calculator alone, automated permits/grid interconnection, AI-only approval of unsafe designs, and storage of raw payment-card data.

Objectives: simplify solar product comparison, provide transparent preliminary estimates, support ordering and assessment workflows, coordinate installation/maintenance, keep auditable records, and protect user information.

1.3 Stakeholders

Stakeholder	Main responsibility
Homeowner	Compare products, estimate needs, order, approve quotes, track work and request maintenance.
Technician	Inspect sites, prepare recommendations, manage assigned jobs and report progress.
Administrator	Manage users, catalog, inventory, assignments, orders, payments and reports.
External providers	Process payments and provide email/SMS and AI services; maps may be optional.
Faculty / development team	Review requirements, design, implementation and project evidence.

2. Requirements

2.1 Requirements Collection Process

The supplied project brief is the initial requirements source. The team should review the workflows with the client, confirm role responsibilities, rank stories using Must/Should/Could/Won't, and record approved changes. Priorities below are provisional recommendations and require client confirmation.

Challenge	Mitigation
Priorities not confirmed	Use provisional priorities and ask the client to approve them.
Incomplete energy/roof inputs	Display assumptions and require technician validation of final design.
Sensitive address/photos	Collect minimum data, restrict access and explain privacy/retention.
Payment provider not selected	Use a provider adapter and confirm the sandbox/provider.
AI may give incorrect advice	Limit informational scope and escalate safety-critical questions.

2.2 Prioritized User Stories

Priority key: P1 = Must have for MVP; P2 = Should have; P3 = Could have. These are provisional recommendations for client review.

US-01 — Authentication and access

As a homeowner, technician, or administrator, I want to register or sign in with role-appropriate access, so that I can use the functions relevant to my role.

Acceptance Criteria

- Required fields and unique email are validated.
- Passwords are stored using secure one-way hashing.
- Errors are clear and do not reveal sensitive account details.
- Restricted pages and API endpoints enforce role permissions.
- Logout invalidates the session/token.

US-02 — Solar capacity estimate

As a homeowner, I want to enter electricity consumption, appliance needs and rooftop area, so that I can obtain a preliminary solar system estimate.

Acceptance Criteria

- User can provide monthly consumption or supported bill/appliance inputs.
- Results show estimated capacity, panel count, generation range and indicative cost.
- Assumptions and limitations are displayed.
- Invalid or incomplete inputs are flagged.
- Estimate can be saved or used to start an assessment request.

US-03 — Browse and compare products

As a homeowner, I want to search, filter and compare solar products, so that I can choose equipment for my needs and budget.

Acceptance Criteria

- Products show available specifications, price, stock and warranty.
- Search and filters update the results.
- Selected products can be compared side by side.
- Unavailable items cannot be ordered.
- Product details are reachable from results.

US-04 — Direct order and checkout

As a homeowner, I want to add products or installation packages to a cart and place a direct order, so that I can purchase a selected solution.

Acceptance Criteria

- Cart totals recalculate when quantities change.
- Stock is checked before confirmation.
- Fees and discounts are itemized.
- User reviews the order before submitting.
- Unique order ID and confirmation are generated; status is trackable.

US-05 — Residential site assessment

As a homeowner, I want to submit energy requirements, rooftop details, location and photos, so that a technician can recommend a suitable solution.

Acceptance Criteria

- Required fields are validated and uploads have type/size limits.
- A request ID and status are provided.
- Authorized staff/assigned technicians can access site details.
- The homeowner can view the next step or appointment.
- Consent and privacy information is visible.

US-06 — Inspection and custom quotation

As an assigned technician, I want to record site findings and prepare an equipment list and quotation, so that the homeowner can make an informed decision.

Acceptance Criteria

- Only assigned technicians can access the job.
- Inspection captures suitability, shading, area and constraints.
- Quotation itemizes equipment, labor/fees, total and validity where applicable.
- Homeowner can approve or reject the quote.
- Decision and timestamp are recorded.

US-07 — Payment and receipt

As a homeowner, I want to pay using supported methods and receive a receipt, so that I have a clear transaction record.

Acceptance Criteria

- Available methods are clearly labelled.
- Online payment is verified server-side with the provider.
- Duplicate callbacks do not create duplicate charges/orders.
- Cash collection, if supported, is recorded with status and responsible staff.
- Receipt includes order ID, amount, date and status.

US-08 — Installation scheduling and tracking

As a homeowner, I want to see appointment details and installation progress, so that I know when the team will arrive and the job status.

Acceptance Criteria

- Appointment and status are visible.
- Status changes are timestamped.
- Allowed states are enforced: Pending, Scheduled, In Progress, Completed, Cancelled.
- Completion requires a checklist/report.
- Configured status changes trigger notifications.

US-09 — Technician job management

As a technician, I want to manage assigned assessments and installation jobs, so that I can plan work and record outcomes.

Acceptance Criteria

- Assigned jobs and schedules are visible.
- Technician can update availability.
- Configured rules prevent conflicting assignments.
- Inspection/progress reports store author and timestamp.
- Completed jobs require an audited correction process for edits.

US-10 — Maintenance and warranty

As a homeowner, I want to request cleaning, repair or maintenance and view service history, so that I can maintain my solar installation.

Acceptance Criteria

- Request captures issue, description, preferred time and installation reference.
- A request ID is generated.
- Staff can assign a technician and update status.
- Authorized users can view warranty and service history.
- Customer can rate completed service.

US-11 — AI solar assistant

As a homeowner, I want to ask solar questions, so that I can understand products, basic sizing and maintenance before contacting a technician.

Acceptance Criteria

- Assistant stays within approved informational scope.
- It does not claim to certify roof/electrical safety.
- Uncertain or safety-critical questions trigger human referral.
- Sensitive account/payment data is not sent unnecessarily to the AI provider.
- Human support remains available.

US-12 — Notifications and support

As a homeowner, I want order, payment, appointment and service updates, so that I do not miss important actions.

Acceptance Criteria

- Notification events are recorded.
- Messages include relevant IDs and next steps.
- Delivery failures are logged and retriable.
- Recent in-app notifications are viewable.
- Support tickets have status and response history.

US-13 — Admin operations and inventory

As an administrator, I want to manage users, products, stock, prices, assessments and orders, so that operational information stays accurate.

Acceptance Criteria

- Admin-only functions enforce RBAC.
- Price and stock changes are validated and audited.
- Workflow transitions are validated.
- Sensitive payment data is not exposed.
- Destructive actions require confirmation or soft deletion.

US-14 — Reports and Analytics

As an administrator/business owner, I want reports on demand, revenue, service progress and satisfaction, so that I can monitor operations.

Acceptance Criteria

- Reports accept a date range.
- Totals reconcile with order/payment records.
- Canceled/refunded transactions are distinguished.
- Only authorized roles can access reports.
- Exports, if implemented, minimize personal data.

2.3 Use Case Diagram

Primary actors are Homeowner, Technician and Administrator. Payment Gateway and AI Service are external systems. The diagram summarizes the main interactions; detailed workflows should be validated with the client.

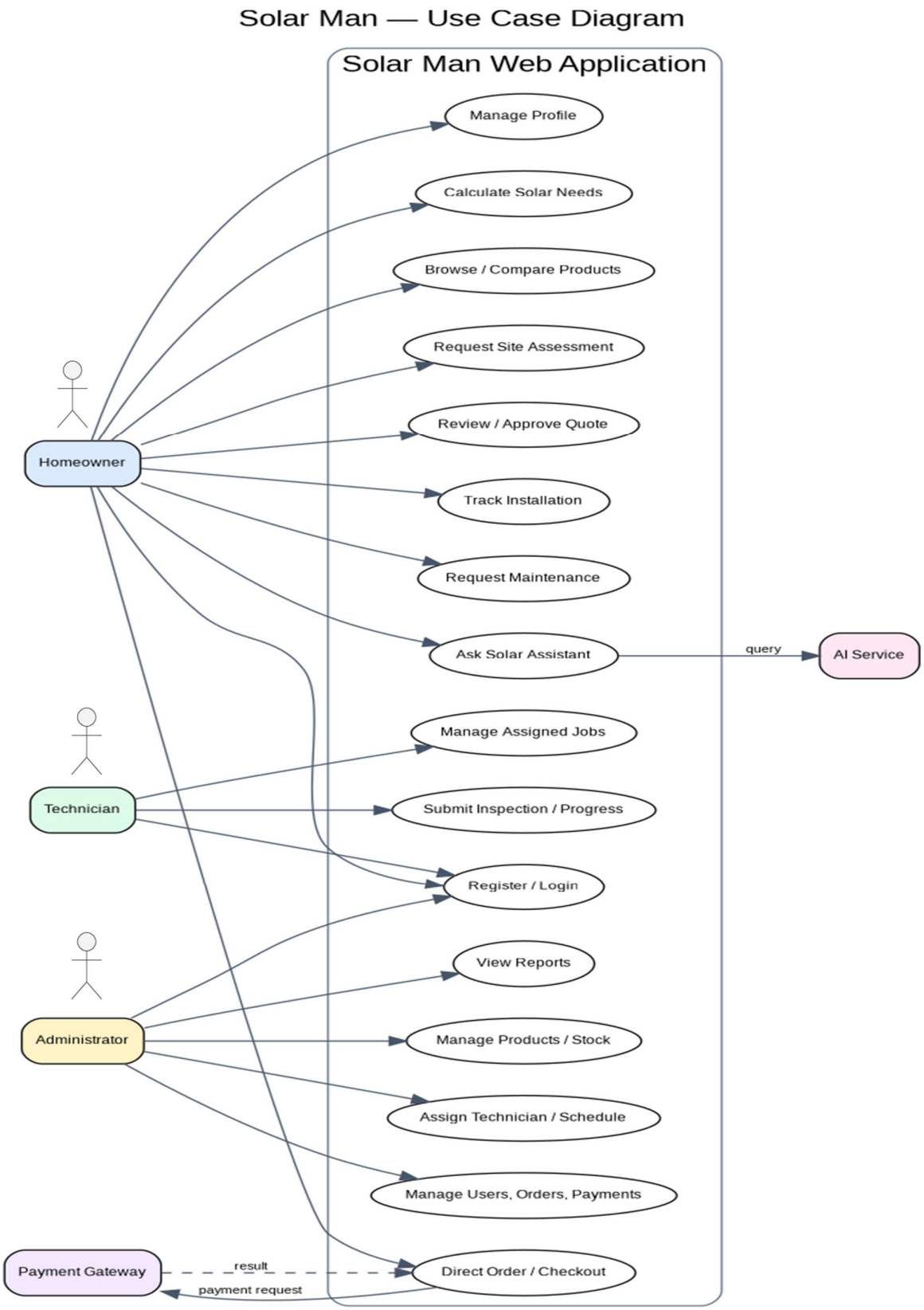


Figure 1: Use Case Diagram

2.4 Brief Use Case Descriptions

Use case	Actor(s)	Main outcome	Story IDs
Register / sign in	All roles	Authenticated session with role permissions	US-01
Calculate solar needs	Homeowner	Estimate and assumptions displayed	US-02
Browse / compare products	Homeowner	Product details and availability displayed	US-03
Direct order / checkout	Homeowner, Payment Provider	Order, payment status and receipt recorded	US-04, US-07
Request assessment	Homeowner, Administrator	Assessment saved and technician assigned	US-05
Inspect / prepare quote	Technician, Homeowner	Inspection and customer quote decision recorded	US-06, US-09
Track installation	Homeowner, Technician, Administrator	Appointment, progress and completion report visible	US-08, US-09
Request maintenance	Homeowner, Technician, Administrator	Service request and history recorded	US-10
Ask AI assistant	Homeowner, AI Service	Bounded answer or human escalation	US-11
Manage operations / reports	Administrator	Validated, audited records and authorized reports	US-12

2.5 Story-to-Use-Case Mapping

Story	Use case / figure
US-01	Register / sign in — Figure 1
US-02	Calculate solar needs — Figure 1
US-03	Browse / compare products — Figure 1
US-04, US-07	Direct order / checkout — Figure 1
US-05	Request assessment — Figure 1

US-06, US-09	Inspect / prepare quote; manage jobs — Figure 1
US-08	Track installation — Figure 1
US-10	Request maintenance — Figure 1
US-11	Ask AI assistant — Figure 1
US-12	Manage operations / reports — Figure 1

Chapter 3: Architecture and Detailed Design

3.1 Overall Architecture

Solar Man uses a modular three-tier web architecture. The frontend handles responsive pages and forms; the application/API layer enforces authentication, authorization, validation and business rules; domain modules implement workflows; the data-access layer manages transactions and audit records. External providers are accessed through adapters. A single modular backend is suitable for the initial student project.

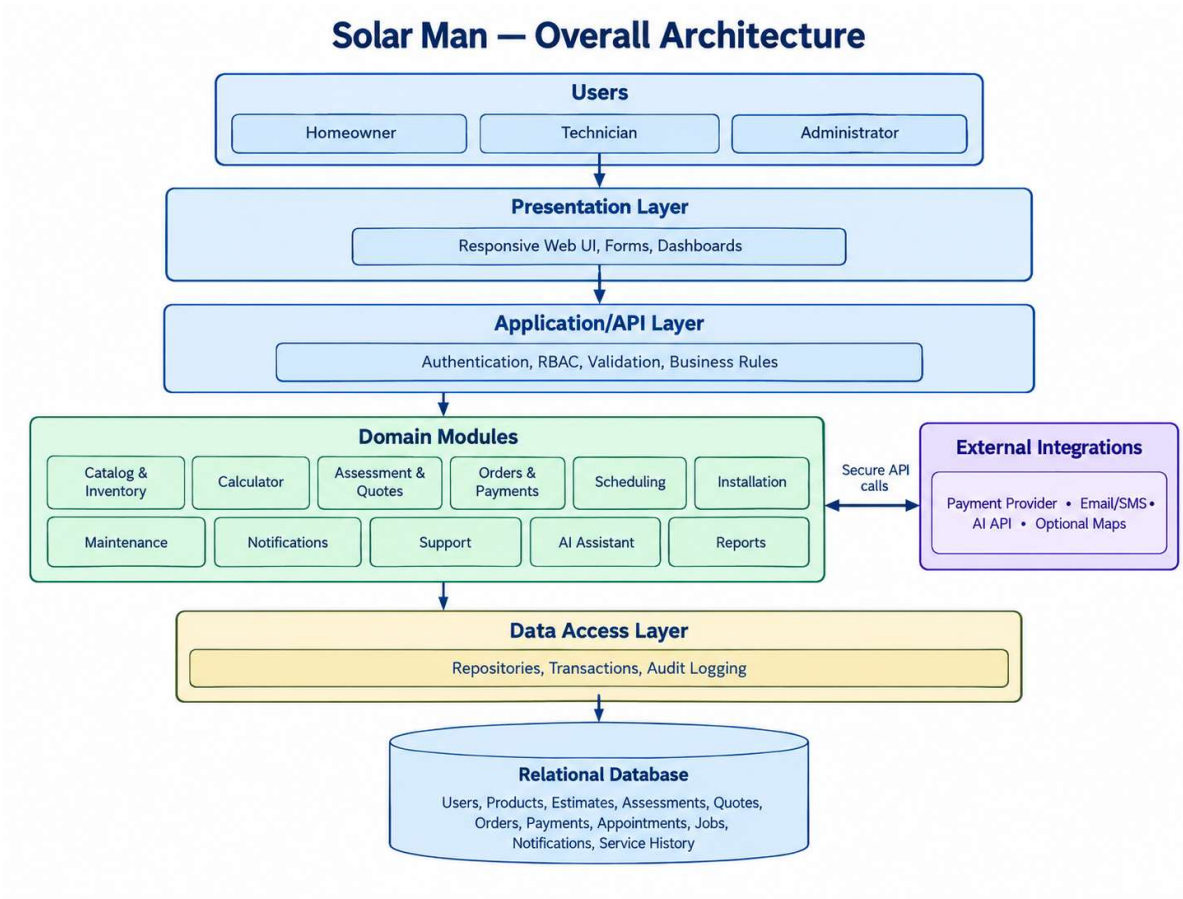


Figure 2: Architecture Design

Layer / module	Responsibility
Presentation	Responsive UI, forms, dashboards and validation messages.
Application/API	Authentication, RBAC, input validation and workflow rules.
Domain modules	Catalog, inventory, calculator, assessment, quotes, orders, payments, scheduling, installation, maintenance, notifications, AI and reports.
Data access / database	Repositories, transactions, audit logging and persistent records.
External integrations	Payment provider, email/SMS, AI API and optional maps.